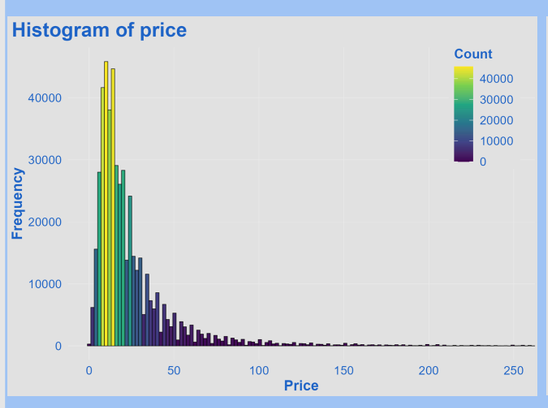
FACULTY OF ECONOMICS AND BUSINESS ADMINISTRATION

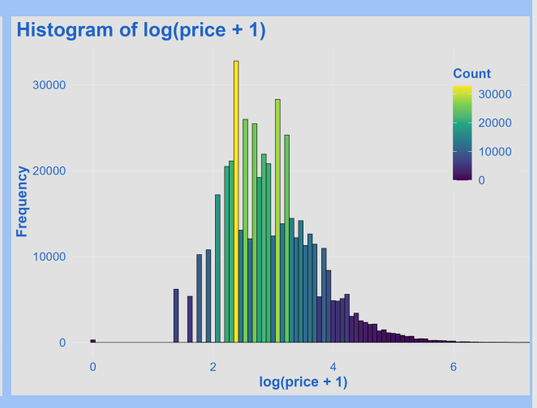
MACHINE LEARNING - GROUP 11
Febe De Smet, Henri Bolliou, Liam Kielemoes & Lucas Minnaert

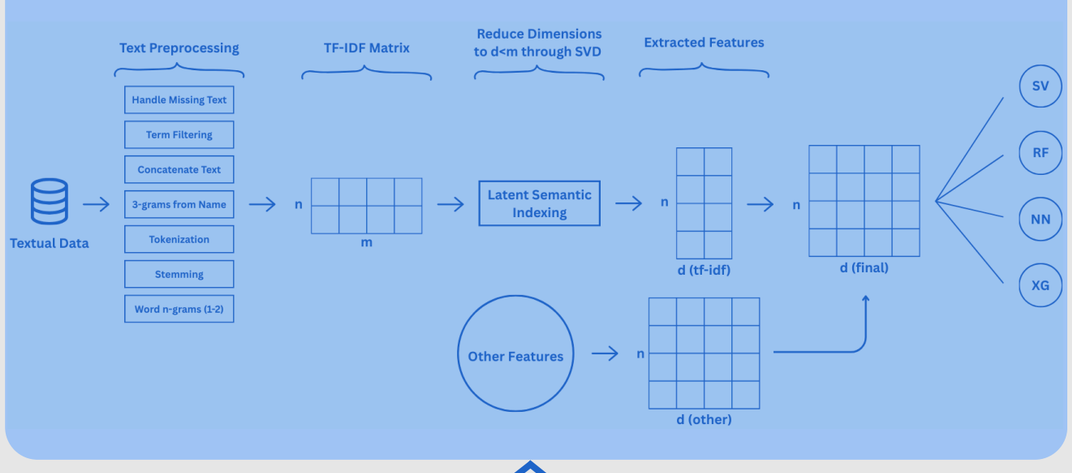
From Text to Tag: Precison Pricing in the Resale Market

1. Data preprocessing

- **Data cleaning:** Lowercase, punctuation, separate categories
- **Smart imputation:** Recovered 1000+ missing brand names from descriptions & title
- **Price normalization:** Stabilized predictions for high-end luxury items to prevent overestimation
- **NLP processing:** Converted unstructured text descriptions into actionable numeric insights → TF-IDF







2. Predictive modelling

2A. XGBoost

- **Precision expert**
- **How it works:** Ensemble of decision trees that learns sequentially
  - Each new tree corrects pricing errors of previous one
- **Data:** Sparse data (TF-IDF)
  - Thrives on full vocabulary without compression
- **Smart feature:** Price-based weighting
  - Pay double attention to high-value items where mistakes are costly
- **Performance**: RMSE 28.22

2B. MLP

- **Multi-layer Perceptron**
- **Context learner**
- **How it works:** Deep neural network with 3 layers
  - Mimics complex pattern recognition
  - Understands hidden connections in descriptions
- **Data**: Dense embeddings (SVD)
  - Compressed text in 150 “concept” dimensions
- **Robustness**: Huber Loss
  - Less sensitive to extreme outliers

  → Stable predictions
- **Performance**: RMSE 26.12

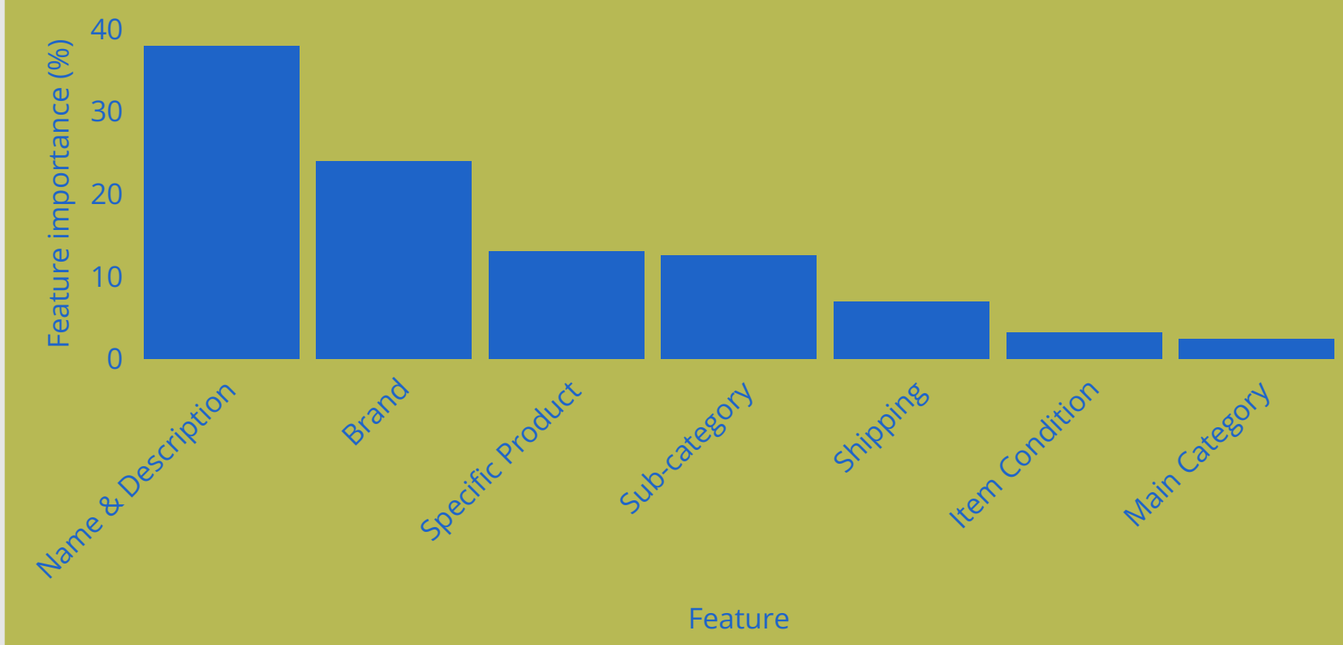
2C. Stack: XGBoost x MLP

- **Logic:** XGBoost excels at specific keyword precision (sparse), MLP masters broad semantic context (dense)

  → Errors are complementary
- **Method:** Generalized Additive Model (GAM)
  - Learns when to trust which model
- **Final result:** Combination outperformed all individual models
- **Best performance: RMSE 25.52**

3. Insights

3A. Feature importances



3B. SHAP values (direction of impact)



3C. Conclusions

1. **Content is king**
   - Name & description → strongest predictor of price
   - Longer names & descriptions → higher valuation
2. **The brand**
   - Brand identity sets price baseline
   - Unknown brand leads to value suppression
3. **Shipping dynamics**
   - Seller pays → low value items
   - Buyer pays → high value items
4. **Categories**
   - Generic categories → weak predictors
   - Specific sub-categories → strong predictors

4. Recommendations

- **Real-time “content strength meter”**
  - Give live feedback on product description & title
  - Nudge sellers to expand titles & descriptions to increase value
- **Close the “brand gap” via AI:**
  - Make it obligatory to select the brand
  - Implement image recognition to auto-suggest brands during upload
- **Mandate granularity:**
  - Force deep categorization by prioritizing specific sub-category selection